

# Liang (Twist) Shan



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🏠 <https://twist-shan.github.io/>

## RESEARCH INTEREST



- Statistical foundations, optimization, and mechanistic interpretability of **Modern Machine Learning**, with a focus on deep learning, reinforcement learning, and generative AI, including language models, diffusion models and AI agents.

## EDUCATION



### Peking University

Beijing, China

- *B.S. in Statistics, School of Mathematical Sciences* Jun. 2024–Jul. 2028 (*Expected*)  
*Second Major B.A. in Economics, National School of Development* Sep. 2025–Jul. 2028 (*Expected*)
  - **Statistics Courses:** Probability Theory, Mathematical Statistics, Applied Stochastic Processes, Statistical Thinking, Stochastic Analysis and Applications.
  - **Computer Science Courses:** Data Structure and Algorithm, Fundamentals of Artificial Intelligence,
  - **Finance and Economics Courses:** Microeconomics, Macroeconomics, Econometrics (Honor), Mathematical Economics, Introduction to Finance, Security Investment, Mathematical Methods in Finance, Modern Quantitative Trading Systems, Topics in Quantitative Finance.

### University of Copenhagen

Copenhagen, Denmark

- *International Exchange Student, Department of SCIENCE* Sep. 2025–Jan. 2026
  - **Mathematics Courses:** Topics in Statistics (MCMC), Advanced Operation Research: Stochastic Programming.
  - **Computer Science Courses:** Probability Machine Learning

### Peking University

Beijing, China

- *B.S. in Physics, School of Earth and Space Sciences* Sep. 2023–Jun. 2024

## RESEARCH EXPERIENCES



### Mechanistic Interpretability of Latent Reasoning in DLMs

University of California, Berkeley

- *Advisor: Prof. Song Mei, UC Berkeley* Aug. 2026 – Present
  - Designed tasks and interpretability pipeline to find out the reasoning mechanics in Diffusion Language Models between different architectures and decoding orders.

### Pareto Frontier Model Identification from Pairwise Preferences

University of Pennsylvania

- *Advisor: Prof. Zhimei Ren, UPenn* May. 2026 – Present
  - Developed an adaptive vector-Borda elimination algorithm for fixed-confidence Pareto frontier identification, with exact recovery and instance-dependent upper and minimax lower bounds on sample complexity.
  - Built regularized-MLE baselines, synthetic benchmarks, and robust preprocessing and replay pipelines for multi-objective evaluation on Real-world Arena data.

### Mechanistic Interpretability of Chain-of-Thought Reasoning

University of Wisconsin–Madison

- *Advisor: Prof. Yiqiao Zhong, UW–Madison* Apr. 2026 – Present
  - Developed controlled synthetic and realistic counting tasks, characterizing how chain-of-thought shifts computation from one-shot broad retrieval to iterative targeted retrieval and counter updates.
  - Built a causal interpretability pipeline using probing, hidden-state steering, position-local head ablation, and activation patching to localize retrieval heads and internal count representations.

## SELECTED COURSE PROJECTS



### LLM Applications: WeChat Content Agent

Peking University

- *Advisor: Prof. Shanghang Zhang, PKU* Apr. 2025 – Jun. 2025
  - Built a Dify-based LLM agent that automated content generation, responsive HTML/CSS typesetting, and draft publishing through Python and WeChat APIs.
  - Integrated RAG and web-retrieval tools to support source-grounded writing and dynamic content collection.

## Empirical Asset Pricing and Portfolio Management

Peking University

Mar. 2025 – Jun. 2025

▪ Advisor: Prof. Hai Huang, PKU

- Analyzed Chinese A-share data and constructed dynamically rebalanced portfolios using mean–variance optimization and fundamental signals.
- Evaluated mutual-fund timing and stock-selection performance using the Treynor–Mazuy and Henriksson–Merton models.

[GitHub](#)

## Quantitative Trading Backtesting Framework

Peking University

Sep. 2024 – Jan. 2025

▪ Advisor: Dr. Changhao Jiang, PKU

- Developed a modular Python backtesting framework that decoupled signal generation, order execution, and performance evaluation.
- Implemented and benchmarked momentum, mean-reversion, cross-sectional, and time-series strategies using vectorized data processing.

[GitHub](#)

## SELECTED HONORS AND AWARDS



### Yu Hao Dream Scholarship

Peking University

May. 2026

▪ Scholarship supporting international exchange program

## SKILLS & OTHERS



- **Computer Skills:** Python (&PyTorch), R, C++ (Elementary), L<sup>A</sup>T<sub>E</sub>X
- **Languages:** Mandarin Chinese (Native), English (Fluent)
- **GRE General Test:** 325/340 (Verbal 155, Quantitative 170)